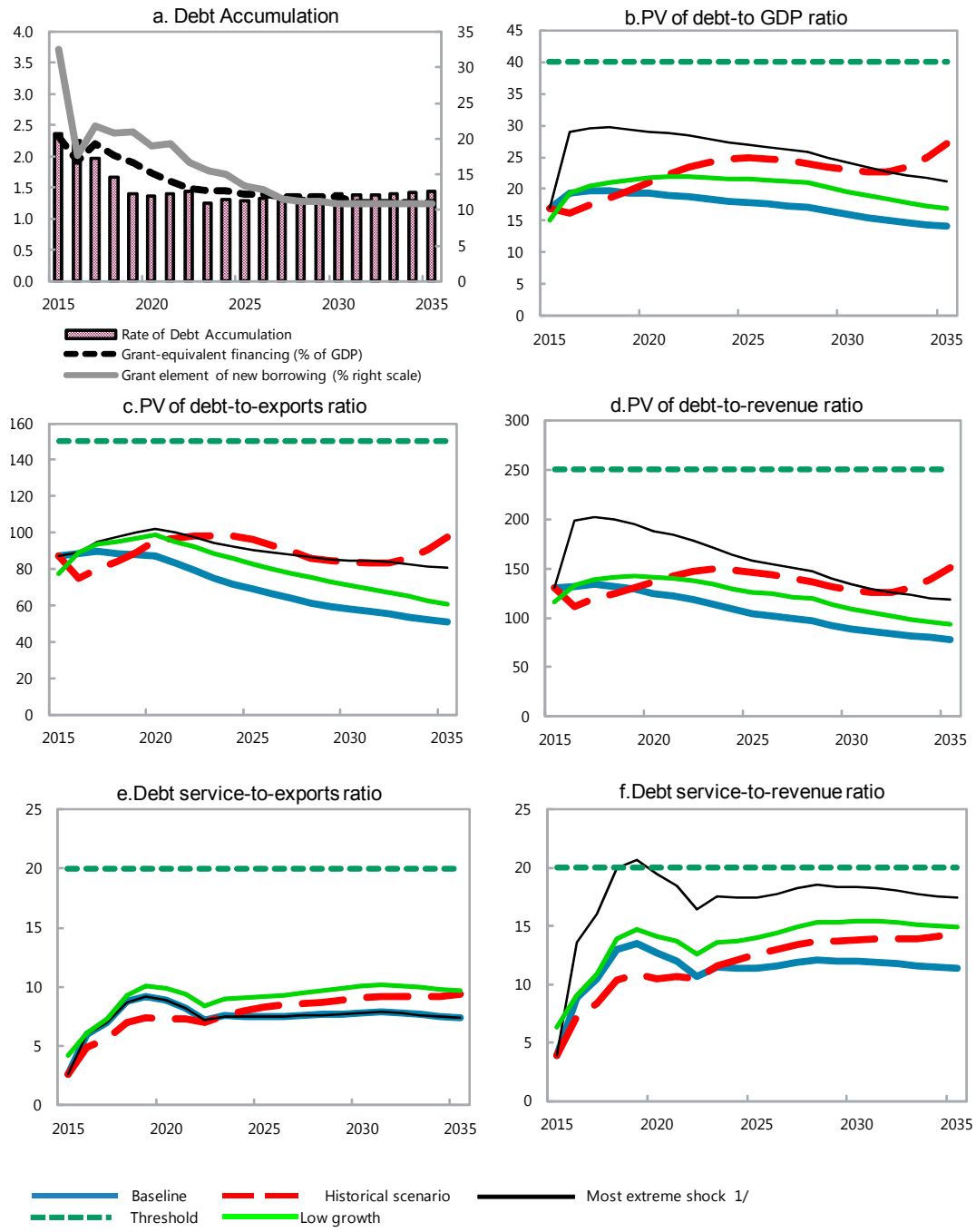
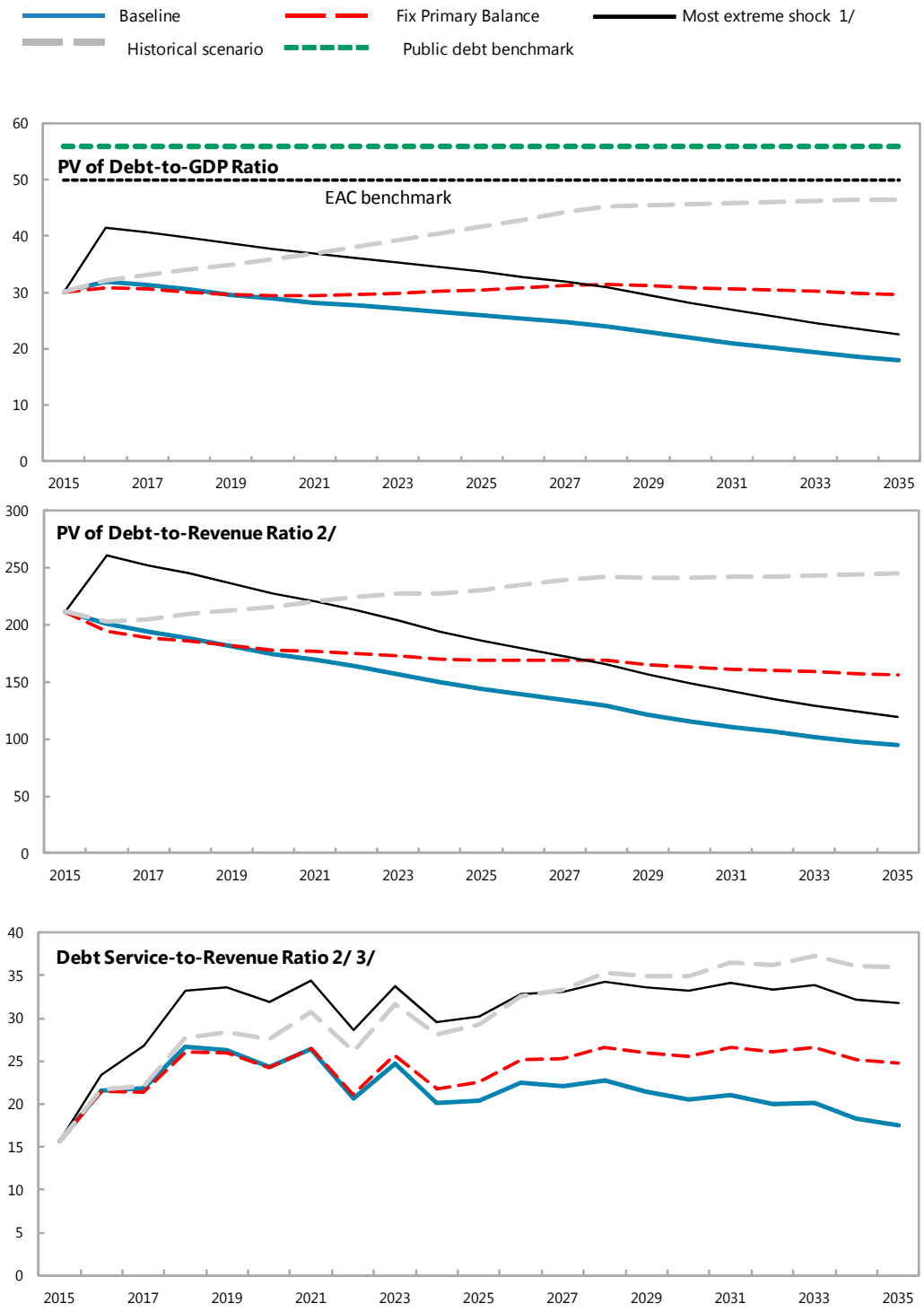
Figure 4. Tanzania: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2015–2035 1/



Sources: Country authorities; and staff estimates and projections.

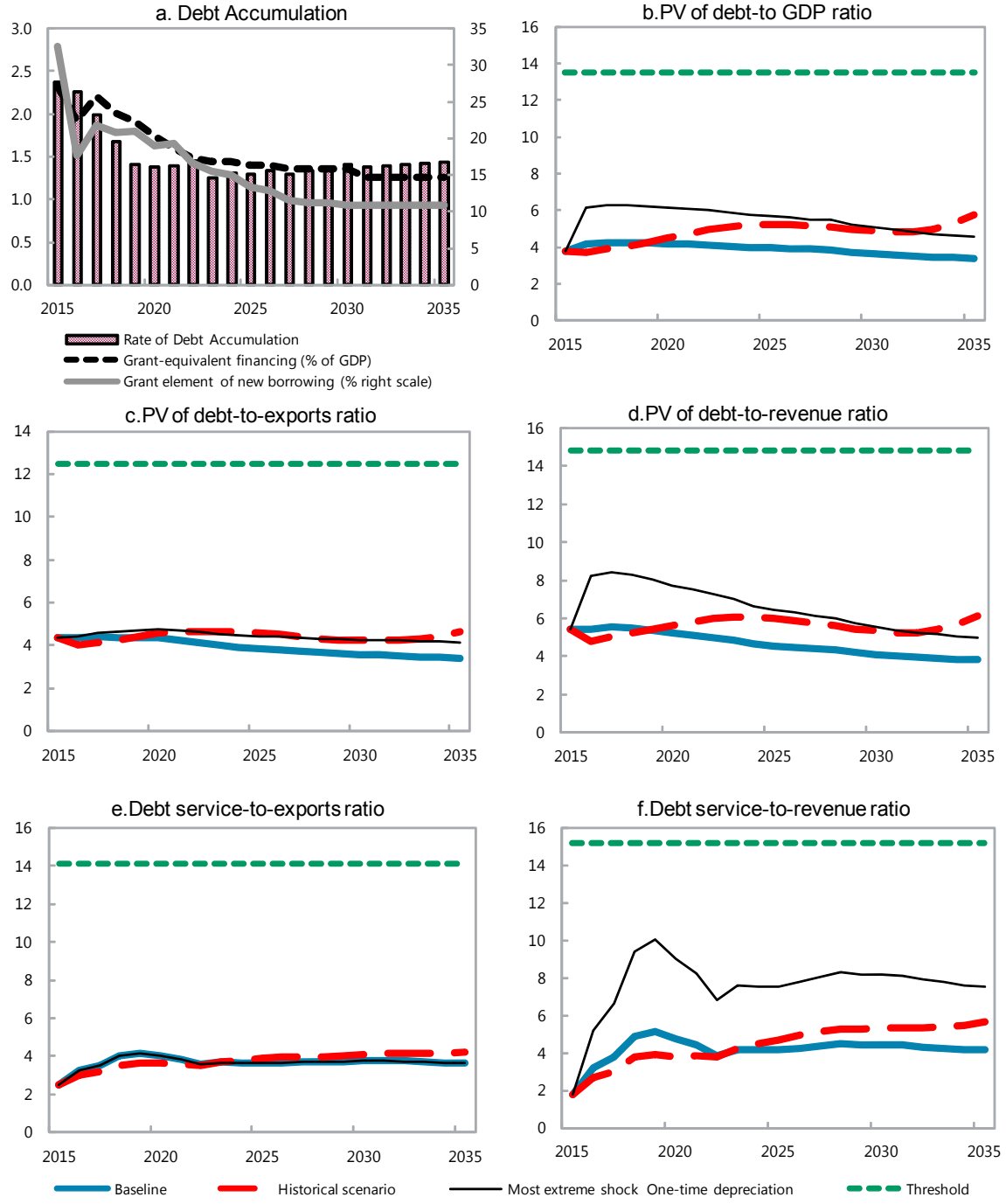
1/ The most extreme stress test is the test that yields the highest ratio on or before 2025. In figure b. it corresponds to a One-time depreciation shock; in c. to a Terms shock; in d. to a One-time depreciation shock; in e. to a One-time depreciation shock and in figure f. to a One-time depreciation shock

Figure 5. Tanzania: Indicators of Public Debt Under Alternative Scenarios, 2015–2035 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2025.
2/ Revenues are defined inclusive of grants.
3/ The volatile profile of debt service is due to the projected armotization for public domestic debt.

Figure 6. Tanzania: Probability of Debt Distress of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2015–2035 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2025. In figure b. it corresponds to a One-time depreciation shock; in c. to a Terms shock; in d. to a One-time depreciation shock; in e. to a One-time depreciation shock and in figure f. to a One-time depreciation shock

Table 6. Tanzania: External Debt Sustainability Framework, Baseline Scenario, 2012–2035 1/

(In percent of GDP, unless otherwise indicated)

	Actual			Historical 6/ Average	Standard 6/ Deviation	Projections									
	2012	2013	2014			2015	2016	2017	2018	2019	2020	2015-2020 Average	2025	2035	2021-2035 Average
External debt (nominal) 1/	**24.5**	**26.5**	**24.5**			**31.7**	**35.0**	**35.2**	**34.9**	**34.5**	**34.2**		**31.3**	**24.5**	
of which: public and publicly guaranteed (PPG)	20.7	22.9	21.3			25.8	28.5	28.6	28.3	27.8	27.4		24.1	18.1	
Change in external debt	-0.8	2.0	-2.0			7.2	3.3	0.2	-0.2	-0.4	-0.3		-0.6	-0.5	
Identified net debt-creating flows	6.7	2.3	3.6			2.6	1.9	1.4	1.3	1.0	0.7		1.5	-0.2	
Non-interest current account deficit	**12.9**	**10.2**	**10.4**	**8.2**	**2.5**	**8.3**	**7.7**	**7.3**	**7.0**	**6.6**	**6.3**		**6.7**	**4.4**	6.5
Deficit in balance of goods and services	13.8	10.8	10.9			8.5	7.3	7.1	6.8	6.3	6.1		6.4	3.7	
Exports	22.3	19.9	19.1			19.4	21.7	21.8	22.1	22.1	22.0		25.9	27.7	
Imports	36.1	30.7	30.0			27.9	29.0	28.9	28.9	28.3	28.1		32.3	31.4	
Net current transfers (negative = inflow)	-2.6	-1.9	-1.6	-2.8	0.7	-1.2	-0.9	-1.3	-1.2	-1.1	-1.2		-0.9	-0.4	-0.8
of which: official	-1.7	-1.2	-0.9			-0.6	-0.1	-0.4	-0.4	-0.3	-0.3		-0.2	-0.1	
Other current account flows (negative = net inflow)	1.6	1.3	1.1			1.0	1.4	1.4	1.4	1.4	1.4		1.2	1.2	
Net FDI (negative = inflow)	**-4.2**	**-4.6**	**-4.4**	**-3.9**	**0.7**	**-4.5**	**-4.0**	**-4.2**	**-4.1**	**-4.0**	**-4.0**		**-4.0**	**-4.0**	-4.0
Endogenous debt dynamics 2/	**-2.0**	**-3.3**	**-2.3**			**-1.2**	**-1.8**	**-1.6**	**-1.6**	**-1.5**	**-1.5**		**-1.2**	**-0.7**	
Contribution from nominal interest rate	0.2	0.3	0.3			0.4	0.6	0.6	0.7	0.7	0.7		0.7	0.8	
Contribution from real GDP growth	-1.5	-1.3	-1.7			-1.7	-2.4	-2.3	-2.2	-2.2	-2.2		-1.9	-1.4	
Contribution from price and exchange rate changes	-0.7	-2.2	-1.0			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**-7.5**	**-0.3**	**-5.7**			**4.6**	**1.4**	**-1.2**	**-1.6**	**-1.4**	**-1.0**		**-2.1**	**-0.3**	
of which: exceptional financing	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	16.7			22.8	25.7	26.2	26.3	26.2	26.1		25.0	20.5	
In percent of exports	...	...	87.4			117.2	118.5	120.2	118.9	118.6	118.5		96.7	73.9	
PV of PPG external debt	**...**	**...**	**13.5**			**16.9**	**19.3**	**19.7**	**19.7**	**19.4**	**19.3**		**17.8**	**14.1**	
In percent of exports	**...**	**...**	**70.6**			**87.0**	**88.8**	**90.1**	**88.9**	**88.1**	**87.4**		**68.9**	**50.9**	
In percent of government revenues	**...**	**...**	**99.5**			**130.3**	**132.0**	**133.9**	**132.2**	**128.9**	**124.7**		**104.8**	**78.3**	
Debt service-to-exports ratio (in percent)	**3.6**	**3.0**	**3.2**			**4.0**	**8.7**	**9.6**	**11.3**	**11.9**	**11.6**		**9.9**	**9.4**	
PPG debt service-to-exports ratio (in percent)	**1.1**	**1.9**	**2.2**			**2.6**	**6.0**	**7.0**	**8.7**	**9.2**	**8.9**		**7.5**	**7.4**	
PPG debt service-to-revenue ratio (in percent)	**1.9**	**3.0**	**3.1**			**3.9**	**8.9**	**10.4**	**13.0**	**13.5**	**12.7**		**11.4**	**11.4**	
Total gross financing need (Billions of U.S. dollars)	3.4	2.6	3.1			2.2	2.5	2.4	2.8	2.9	2.9		4.9	8.2	
Non-interest current account deficit that stabilizes debt ratio	13.7	8.2	12.4			1.1	4.4	7.1	7.2	7.0	6.6		7.3	4.9	
Key macroeconomic assumptions															
Real GDP growth (in percent)	6.5	6.2	7.1	6.4	0.6	7.0	7.0	7.0	6.9	6.9	6.9	6.9	6.5	6.5	6.5
GDP deflator in US dollar terms (change in percent)	2.8	10.0	3.8	5.6	5.4	-3.9	-14.1	1.0	1.4	1.6	1.2	-2.1	2.4	5.4	3.8
Effective interest rate (percent) 5/	0.9	1.4	1.3	1.0	0.3	1.9	1.8	1.9	2.0	2.1	2.1	2.0	2.5	3.5	2.8
Growth of exports of G&S (US dollar terms, in percent)	13.3	4.4	6.6	14.7	8.3	4.7	2.6	8.6	10.0	8.3	8.0	7.0	11.8	13.0	12.2
Growth of imports of G&S (US dollar terms, in percent)	29.5	-0.6	8.5	17.2	11.8	-4.3	-4.7	7.7	8.5	6.3	7.2	3.5	12.0	10.0	11.3
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	32.5	17.7	21.7	20.8	21.0	18.9	22.1	13.3	10.8	12.7
Government revenues (excluding grants, in percent of GDP)	12.6	12.8	13.5			13.0	14.6	14.7	14.9	15.1	15.4		17.0	18.0	17.3
Aid flows (in Billions of US dollars) 7/	1.2	1.1	1.0			1.1	0.9	1.1	1.1	1.2	1.2		1.4	3.7	
of which: Grants	1.2	1.1	1.0			0.6	0.6	0.7	0.7	0.7	0.7		1.0	2.5	
of which: Concessional loans	0.0	0.0	0.0			0.5	0.4	0.4	0.4	0.5	0.5		0.5	1.2	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			2.3	1.9	2.2	2.0	1.9	1.7		1.4	1.3	1.4
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			50.3	38.3	46.4	43.3	43.1	41.1		37.2	31.8	35.2
Memorandum items:															
Nominal GDP (Billions of US dollars)	35.8	41.9	46.6			47.8	44.0	47.5	51.5	55.9	60.5		92.8	270.5	
Nominal dollar GDP growth	9.4	16.9	11.1			2.8	-8.1	8.0	8.4	8.6	8.2	4.7	9.0	12.3	10.5
PV of PPG external debt (in Billions of US dollars)			6.1			7.2	8.3	9.2	10.0	10.7	11.5		16.3	37.6	
(PVt-PVt-1)/GDPt-1 (in percent)						5.3	2.3	2.0	1.7	1.4	1.4	2.3	1.3	1.4	1.4
Gross workers' remittances (Billions of US dollars)	0.0	0.1	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.1	0.1	
PV of PPG external debt (in percent of GDP + remittances)	...	...	13.5			16.9	19.3	19.6	19.7	19.4	19.2		17.8	14.1	
PV of PPG external debt (in percent of exports + remittances)	...	...	70.3			86.7	88.5	89.8	88.6	87.8	87.1		68.8	50.8	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	2.2			2.6	5.9	7.0	8.7	9.2	8.8		7.5	7.4	

Sources: Tanzanian authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.
2/ Derived as [r - g - ρ(1+g))/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that PV of private sector debt is equivalent to its face value.
5/ Current-year interest payments divided by previous period debt stock.
6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.
7/ Defined as grants, concessional loans, and debt relief.
8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 7. Tanzania: Public Sector Debt Sustainability Framework, Baseline Scenario, 2012–2035
(in percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2012	2013	2014	Average 5/	Standard Deviation 5/	2015	2016	2017	2018	2019	2020	2015-20 Average	2025	2035	2021-35 Average
Public sector debt 1/	27.2	29.7	29.0			38.9	41.2	40.3	39.1	37.9	37.0		32.3	22.0	
of which: foreign-currency denominated	20.7	22.9	21.3			25.8	28.5	28.6	28.3	27.8	27.4		24.1	18.1	
Change in public sector debt	0.0	2.5	-0.7			9.9	2.3	-1.0	-1.2	-1.2	-0.9		-0.9	-0.7	
Identified debt-creating flows	-0.6	2.2	1.4			10.9	2.5	-0.6	-0.5	-0.5	-0.3		-0.5	-0.5	
Primary deficit	2.9	3.8	2.0	2.9	1.1	1.6	2.8	1.3	1.2	1.1	1.1	1.5	0.6	0.6	0.6
Revenue and grants	15.9	15.4	15.7			14.2	15.9	16.2	16.2	16.3	16.6		18.1	19.0	
of which: grants	3.2	2.6	2.1			1.2	1.3	1.5	1.3	1.2	1.2		1.0	0.9	
Primary (noninterest) expenditure	18.8	19.2	17.7			15.8	18.6	17.5	17.4	17.5	17.7		18.7	19.6	
Automatic debt dynamics	-3.5	-2.2	-1.6			2.0	-0.2	-1.9	-1.7	-1.6	-1.4		-1.2	-1.1	
Contribution from interest rate/growth differential	-1.8	-1.3	-1.3			-1.1	-2.2	-2.1	-1.8	-1.7	-1.6		-1.1	-0.5	
of which: contribution from average real interest rate	-0.2	0.3	0.6			0.8	0.4	0.6	0.8	0.8	0.8		0.9	0.9	
of which: contribution from real GDP growth	-1.7	-1.6	-2.0			-1.9	-2.5	-2.7	-2.6	-2.5	-2.4		-2.0	-1.4	
Contribution from real exchange rate depreciation	-1.6	-0.9	-0.3			3.1	2.0	0.2	0.1	0.1	0.2		...	...	
Other identified debt-creating flows	0.0	0.6	1.0			7.3	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			6.2	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.6	1.0			1.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	0.6	0.4	-2.1			-1.0	-0.3	-0.4	-0.7	-0.7	-0.6		-0.3	-0.3	
Other Sustainability Indicators															
PV of public sector debt	...	...	21.2			30.0	31.9	31.3	30.5	29.6	28.9		26.0	18.0	
of which: foreign-currency denominated	...	...	13.5			16.9	19.3	19.7	19.7	19.4	19.3		17.8	14.1	
of which: external	...	...	13.5			16.9	19.3	19.7	19.7	19.4	19.3		17.8	14.1	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	3.9	5.8	3.8			3.8	6.2	4.8	5.5	5.4	5.1		4.3	3.9	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	135.5			211.6	201.3	193.5	187.9	181.1	174.0		144.1	94.6	
PV of public sector debt-to-revenue ratio (in percent)	...	...	156.6			231.4	218.6	213.2	204.6	196.0	187.1		152.9	99.5	
of which: external 3/	...	...	99.5			130.3	132.0	133.9	132.2	128.9	124.7		104.8	78.3	
Debt service-to-revenue and grants ratio (in percent) 4/	6.7	13.3	11.2			15.7	21.5	21.9	26.7	26.2	24.3		20.4	17.5	
Debt service-to-revenue ratio (in percent) 4/	8.4	16.0	13.0			17.1	23.4	24.1	29.1	28.4	26.1		21.6	18.4	
Primary deficit that stabilizes the debt-to-GDP ratio	2.9	1.3	2.7			-8.3	0.5	2.3	2.4	2.3	2.0		1.5	1.3	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	6.5	6.2	7.1	6.4	0.6	7.0	7.0	7.0	6.9	6.9	6.9	6.9	6.5	6.5	6.5
Average nominal interest rate on forex debt (in percent)	0.9	1.5	1.5	0.9	0.4	2.0	1.7	1.9	2.1	2.1	2.1	2.0	2.7	4.0	3.1
Average real interest rate on domestic debt (in percent)	1.3	6.2	9.7	3.5	3.5	10.3	3.5	5.6	7.2	7.8	8.5	7.1	9.8	13.9	11.3
Real exchange rate depreciation (in percent, + indicates depreciation)	-8.2	-4.7	-1.2	-3.0	5.5	15.5	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	11.0	9.1	6.1	9.8	2.1	4.9	5.0	4.7	4.7	4.7	4.3	4.7	5.4	5.4	5.4
Growth of real primary spending (deflated by GDP deflator, in percent)	0.0	0.1	0.0	0.1	0.1	0.0	0.3	0.0	0.1	0.1	0.1	0.1	0.1	0.1	0.1
Grant element of new external borrowing (in percent)	...	...	...	...	...	32.5	17.7	21.7	20.8	21.0	18.9	22.1	13.3	10.8	...

Sources: Country authorities; and staff estimates and projections.
1/ Gross public sector debt covers general government or non-financial public sector.
2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.
3/ Revenues excluding grants.
4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.
5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 8. Tanzania: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2015–2035

(In percent)

	Projections							
	2015	2016	2017	2018	2019	2020	**2025**	2035
PV of debt-to GDP ratio								
Baseline	17	19	20	20	19	19	**18**	14
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	17	16	18	19	20	21	**25**	27
A2. New public sector loans on less favorable terms in 2015-2035 2	17	19	21	22	22	23	**24**	22
A3. Alternative Scenario : Low growth	15	19	20	21	21	22	**21**	17
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	17	19	20	20	20	19	**18**	14
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	17	18	18	18	18	18	**17**	14
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	17	16	17	17	16	16	**15**	12
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	17	18	19	19	19	19	**17**	14
B5. Combination of B1-B4 using one-half standard deviation shocks	17	13	10	10	10	10	**12**	11
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	17	29	30	30	29	29	**27**	21
PV of debt-to-exports ratio								
Baseline	87	89	90	89	88	87	**69**	51
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	87	75	80	84	89	96	**96**	98
A2. New public sector loans on less favorable terms in 2015-2035 2	87	89	95	97	100	102	**91**	81
A3. Alternative Scenario : Low growth	78	89	94	95	97	99	**83**	61
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	87	87	88	87	86	86	**68**	50
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	87	80	82	81	81	80	**65**	49
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	87	87	88	87	86	86	**68**	50
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	87	85	87	86	85	85	**67**	50
B5. Combination of B1-B4 using one-half standard deviation shocks	87	65	50	51	51	52	**49**	44
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	87	87	88	87	86	86	**68**	50
PV of debt-to-revenue ratio								
Baseline	130	132	134	132	129	125	**105**	78
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	130	111	119	125	131	136	**146**	151
A2. New public sector loans on less favorable terms in 2015-2035 2	130	133	141	145	146	146	**138**	124
A3. Alternative Scenario : Low growth	116	133	139	142	142	141	**126**	94
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	130	131	134	133	129	125	**105**	79
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	130	124	124	123	120	117	**100**	76
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	130	111	113	112	109	106	**89**	66
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	130	126	129	128	124	121	**102**	77
B5. Combination of B1-B4 using one-half standard deviation shocks	130	87	68	69	68	68	**68**	62
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	130	198	202	199	194	188	**158**	118

Table 8. Tanzania: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2015–2035 (continued)

(In percent)

Debt service-to-exports ratio								
Baseline	3	6	7	9	9	9	**7**	7
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	3	5	6	7	7	7	**8**	9
A2. New public sector loans on less favorable terms in 2015-2035 2	3	6	6	8	8	8	**7**	10
A3. Alternative Scenario : Low growth	4	6	7	9	10	10	**9**	10
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	3	6	7	9	9	9	**7**	7
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	3	6	7	8	9	8	**7**	7
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	3	6	7	9	9	9	**7**	7
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	3	6	7	9	9	9	**7**	7
B5. Combination of B1-B4 using one-half standard deviation shocks	3	6	6	7	7	6	**5**	6
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	3	6	7	9	9	9	**7**	7
Debt service-to-revenue ratio								
Baseline	4	9	10	13	13	13	**11**	11
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	4	7	8	10	11	10	**13**	14
A2. New public sector loans on less favorable terms in 2015-2035 2	4	9	10	12	11	11	**11**	15
A3. Alternative Scenario : Low growth	6	9	11	14	15	14	**14**	15
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	4	9	11	13	14	13	**12**	12
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	4	9	10	13	13	12	**11**	11
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	4	8	9	11	12	11	**10**	10
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	4	9	10	13	13	12	**11**	11
B5. Combination of B1-B4 using one-half standard deviation shocks	4	7	8	9	10	8	**7**	9
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	4	14	16	20	21	19	**17**	17.39
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	6	6	6	6	6	6	**6**	6

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.

2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.

3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).

4/ Includes official and private transfers and FDI.

5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.

6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 9. Tanzania: Sensitivity Analysis for Key Indicators of Public Debt 2015–2035

	Projections							
	2015	2016	2017	2018	2019	2020	2025	2035
PV of Debt-to-GDP Ratio								
Baseline	30	32	31	30	30	29	26	18
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	30	32	33	34	35	36	42	47
A2. Primary balance is unchanged from 2015	30	31	31	30	30	29	30	30
A3. Permanently lower GDP growth 1/	30	32	31	31	30	29	27	22
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2016-2017	30	32	33	32	31	31	30	23
B2. Primary balance is at historical average minus one standard deviations in 2016-2017	30	33	35	34	33	32	29	20
B3. Combination of B1-B2 using one half standard deviation shocks	30	33	34	34	33	33	30	23
B4. One-time 30 percent real depreciation in 2016	30	40	39	38	37	36	33	26
B5. 10 percent of GDP increase in other debt-creating flows in 2016	30	41	41	40	39	38	34	23
PV of Debt-to-Revenue Ratio 2/								
Baseline	212	201	194	188	181	174	144	95
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	212	203	205	210	213	216	231	245
A2. Primary balance is unchanged from 2015	212	194	189	186	182	178	169	156
A3. Permanently lower GDP growth 1/	212	202	194	189	183	177	151	116
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2016-2017	212	204	200	197	192	186	164	123
B2. Primary balance is at historical average minus one standard deviations in 2016-2017	212	209	216	210	203	195	161	104
B3. Combination of B1-B2 using one half standard deviation shocks	212	207	212	208	202	196	169	122
B4. One-time 30 percent real depreciation in 2016	212	255	243	234	225	216	185	136
B5. 10 percent of GDP increase in other debt-creating flows in 2016	212	261	251	245	236	227	186	119
Debt Service-to-Revenue Ratio 2/								
Baseline	16	22	22	27	26	24	20	17
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	16	22	22	28	28	28	29	36
A2. Primary balance is unchanged from 2015	16	22	21	26	26	24	22	25
A3. Permanently lower GDP growth 1/	16	22	22	27	26	25	21	20
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2016-2017	16	22	22	28	27	25	23	21
B2. Primary balance is at historical average minus one standard deviations in 2016-2017	16	22	22	29	29	27	22	19
B3. Combination of B1-B2 using one half standard deviation shocks	16	22	22	28	28	27	23	21
B4. One-time 30 percent real depreciation in 2016	16	23	27	33	34	32	30	32
B5. 10 percent of GDP increase in other debt-creating flows in 2016	16	22	26	33	32	30	26	21

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.

Table 10. Tanzania: Summary Table on External Borrowing Program

PPG external debt contracted Year 2015/16	Volume of new debt (USD million) 1/	Present value of new debt (USD million) 1/
Sources of debt financing	**1254.8**	**985.3**
Concessional debt, of which 2/	429.7	194.4
Multilateral debt	395.7	177.2
Bilateral debt	34	17.2
Non- Concessional debt, of which 2/	825.1	790.9
Semi-concessional debt 3/	325.1	290.9
Commercial-term debt 4/	500	500
Use of debt financing	**1254.8**	**985.3**
Infrastructure 5/	552.3	433.5
Social Spending	185.7	144.8
Budget financing	516.8	406.93
Memorandum items		
Indicative projections		
Year 2: FY 2016/17	1544	231.6
Year 3: FY 2017/18	1712.3	256.8

1/ Contracting and guaranteeing of new debt. The present value of debt is calculated using the terms of individual loans and applying the 5 percent program discount rate.
2/ Debt with a grant element that exceeds the minimum threshold (35 percent).
3/ Debt with a positive grant element which does not meet the minimum grant element.
4/ Debt without a positive grant element. For commercial debt, the present value would be defined as the nominal/face value.
5/ Infrastructure projects include airport, roads, water supply, and power sector.